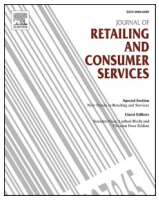
Journal of Retailing and Consumer Services 72 (2023) 103242



Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser





Trust me, I'm an influencer! - Causal recipes for customer trust in artificial intelligence influencers in the retail industry

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ARTICLE INFO

Keywords:
Influencer marketing
Artificial intelligence
Source attractiveness
Source credibility
Congruencies
Trust
fsQCA

ABSTRACT

Prior examinations of relationship development and the leverage of trust among artificial intelligence (AI) influencers and followers have been few. This study employed complexity theory to understand the main causal recipes that can lead to high trust in AI influencers. Asymmetrical fuzzy set qualitative comparative analysis (fsQCA) was used to explore the recipes that can drive high customer trust. Data were collected from 683 consumers who are familiar with AI influencers in Saudi Arabia. Our findings indicated that no single factor is sufficient to drive trust in influencers, but five causal recipes were explored for their power to secure high levels of trust in AI influencers. The findings revealed that a configuration of source attractiveness (i.e., physical attractiveness, homophily), source credibility (i.e., authenticity, expertise) and congruences (i.e., influencer, product, consumer) act as driver of consumers' trust in an AI influencer. These results are useful for practitioners since they provide new methods for boosting trust in AI influencers.

1. Introduction

Social media advertising has seen a dramatic transformation during the past decade, marked by a trend toward influencer marketing (Vrontis et al., 2021a,b). In the year 2020, the influencer marketing market was worth $9.7 billion, and this number is projected to increase to $15 billion by 2023 (Statistica, 2022). The market was only $500,000 in 2015 but has risen at a rate of over 50% annually (Statistica, 2022). As more individuals resorted to online platforms for news, support and recreation during the COVID19 pandemic, influencer marketing grew in popularity (Seçilmiş et al., 2022; Taylor, 2020). Thus, influencer marketing has evolved into an integral aspect of digital marketing strategy.

Marketers often refer to social media influencers as the "new brand" because of their very heavy impact on consumer behaviour (Femenia--Serra et al., 2022; Pick, 2020). However, a new category of online celebrity has arisen: those who use artificial intelligence (AI) to exert a sway in cyberspace (Sands et al., 2022). The World Health Organization's use of AI influencer Knox Frost in the worst days of the SARS and COVID-19 pandemics is a testament to the growing stature of these figures. With over a million Instagram followers, Frost was recruited to spread PSAs like lockdown regulations and the need for personal hygiene (Abdelmoety et al., 2022; Farivar and Wang, 2022; Koay et al., 2022). AI influencers are nothing new; in 2018, French luxury brand Balmain replaced the Kardashians with three AIs, Shudu, Margot and Xhi. Balmain's press release claimed that these personalities best exemplified the brand's commitment to diversity and tolerance (Aboul-Dahab et al., 2021; Kim and Kim, 2021; Barta et al., 2022). A number of well-known companies have partnered with AI influencers, including KFC, LVMH, Mini, Netflix, Nike and Samsung (Agag, 2019; Koay et al., 2022).

Influencers in the field of AI can take several forms. For instance, the AI Lil Miquela (@lilmiquela) was developed in 2016 and she already has over 3 million Instagram followers. Another AI influencer, Shudu Gramm (@shudu.gramm), bills itself as "the world's first digital supermodel" and has formed partnerships with major publications and brands such as the UK sportswear label, Ellesse. While Lil Miquela's influencer account is carefully crafted to make her seem like a skilled "20-something-year-old", someone to aspire to and with a burgeoning music career, Shudu Gramm's is touted instead as an editorial "model" or something to look at (Sands et al., 2022). However, just like humans, AIs construct identities and they engage with brands in remarkably human ways.

Long-term collaborations between companies and influencers are emerging as a result of the rapid development of influencer marketing in today's increasingly competitive social media landscape (Agag et al., 2022; Kim and Kim, 2021; Martínez-López et al., 2020a,b). It is possible to view influencer marketing as a multi-layered form of relationship marketing that involves relationships between influencers and

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https://doi.org/10.1016/j.jretconser.2022.103242
Received 12 November 2022; Received in revised form 10 December 2022; Accepted 24 December 2022
Available online 30 December 2022
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consumers, influencers and brands and brands and consumers. Marketers have recognised the value of continuously weaving their marketing messages into the narratives of influencers gradually over time in order to increase the engagement of followers (Acikgoz and Burnaz, 2021; Farivar et al., 2021; Leung et al., 2022). To be more specific, influencers spread the word about a business by capitalising on the rapport and credibility with their audience that they have so far built. This is why it is so important for such relationships to leverage trust, the cornerstone of any successful leader-follower dynamic (Leite and Baptista, 2022; Trivedi and Sama, 2020). According to a survey, 92% of social media users trust influencers more than traditional marketing channels, demonstrating the significance of trust in influencer marketing (Agag et al., 2020; Jin et al., 2019; Pop et al., 2022). According to Ipsos's 2019 global consumer poll, 46% of respondents do not trust traditional media outlets, whether newspapers, magazines, TV, or radio, while 63% are more likely to follow the advice of social media influencers than they are to pay attention to corporate advertisements (Agag and Colmekcioglu, 2020; Pozharliev et al., 2022; Zhou et al., 2021). In order to thrive in this new landscape, social media marketers need to acquire new marketing skills and knowledge, specifically related to the cultivation and maintenance of trust between online opinion leaders and their virtual constituencies.

Because individuals place a high value on the physical appearance of the trustee when forming a bond of trust (Agag and Eid, 2019; Chong et al., 2022), the nature of the embodiment is forecast to play a significant role in the formation of bonds of trust between humans and AI influencers. To date, most research on AI advocates has concentrated on the elements that affect how often and how well they are used in the average person's daily life. Moriuchi (2019) looked into the TAM components and demonstrated how voice technology affects user loyalty and involvement. Guzman (2019) used Cognitive Appraisal Theory to investigate the primary motivators for integrating AI into service provision. Finally, Chong et al. (2022) investigated people's trust in AI influencers. However, few studies have investigated what determines users' trust in AI influencers, favouring instead either an information systems viewpoint (Nasirian et al., 2020) or a social reaction approach (Agag and El-Masry, 2016; Foehr and Germelmann, 2020). However, a thorough comprehension of the primary elements influencing users' trust in AI influencers has yet to be created. Moreover, existing technological acceptance theories provide the backbone of studies that have hitherto analysed customers' trust in AI influencers. Using the Technology Acceptance Model (TAM), Sundar et al. (2016) conducted a study on the topic of people's willingness to adopt AI robots. The Unified Theory of Acceptance and Use of Technology (UTAUT) was applied in Fritz et al. (2016) research on the likelihood that seniors will adopt AI-powered home monitoring systems. Traditional technology acceptance models may partially explain the mechanism by which users intend to use AI devices; however, these models were designed to study the spread of dumb technologies, such as self-service ones in service delivery (Agag and El-Masry, 2017; Lee and Cranage, 2018), whereas AI influencers are endowed with artificial intelligence that is indistinguishable from that of a human. Therefore, the ease-of-use construct found in all technology acceptance models as a core construct is irrelevant to the study of customers' readiness to accept the employment of AI influencers during a service encounter because these devices do not require users to learn how to operate them (Lu et al., 2019). Despite the increasing attention, individuals' trust in AI has not been understood well enough to avoid faulty decisions to either accept or reject AI influencers.

By integrating previous research on influencer marketing and complexity theory, our research aims to address this gap' it explores the main antecedents of customers' trust in AI influencers. Our study employs complexity theory to explore the main configurations among variables that may drive consumers' trust toward AI influencers. in detail, this study aims to identify how source attractiveness (i.e., physical attractiveness, homophily), source credibility (i.e., expertise, authenticity), and congruence (i.e., influencer, customer and product) combine together to form patterns that explain high trust in AI influencers. To this end, our paper seeks to address the following research question: "What configurations of source attractiveness, congruences, and source credibility lead to increased trust in AI influencers"?

This paper responds to Lu et al. (2020)'s calls to further examine customers' trust in AI influencers. Therefore, our study contributes to the previous studies in several ways. First, the findings of this study provide empirical evidence of the main drivers of customers' trust in AI influencers, indicating the role of source attractiveness, source credibility and congruences in developing trust in AI influencers (Chong et al., 2022). Second, it indicates the critical role of congruences elements, namely congruence between consumer and AI influencer, congruence between AI influencer and product and congruence between consumer and product, as unique drivers for developing individuals' trust in AI influencers (Agag et al., 2019; Belanche et al., 2021). While prior research suggested that congruence attributes enhance customers' trust in the online environment, this effect has rarely been empirically explored with reference to customers–AI influencer interactions. Finally, our paper demonstrates that causal recipes lead to high levels of trust in an AI influencer, while offering support for using complexity theory to understand how different configurations can achieve customers' trust in an AI influencer; i.e. through the identification of antecedents incorporating source attractiveness (i.e., physical attractiveness, homophily) and source credibility (i.e., expertise, authenticity), while including congruence between influencer, customers and product.

The next section reviews the literature review on this study. The third section presents the conceptual framework and hypotheses development. Data collection and analysis are demonstrated in section four. The findings are shown in section five and the discussion and conclusion are contained in section six. Section seven lists the limitations of the current study and suggests some directions for further research.

2. Literature review and propositions development

2.1. Complexity theory

Prior studies employed complexity theory in different contexts, such as hospitality and tourism (e.g., Manosuthi et al., 2022; Shi et al., 2022), natural and social science (e.g., Alyahya et al., 2023; Gligor et al., 2022), and marketing (e.g., Alsuwaidi et al., 2022; Li et al., 2022), which offer a deeper insight into configurations of causal recipes in achieving outcome conditions. Agag et al. (2020) utilised complexity theory to explain factors affecting travellers' willingness to pay more for green travel products. Chuah et al. (2021) employed complexity theory to unveil the complexity of consumers' intention to use service robots. Thus, our study views complexity theory as a necessary and sufficient theoretical base on configurational models for driving customers trust toward AI influencer.

In complexity theory, researchers look for recurrent themes that shed light on the interplay between underlying causes and their effects. Many principles join to form the basis of complexity theory (Alyahya et al., 2022; Woodside et al., 2015). To accurately forecast the desired outcome takes more than one component (though further factors may also be necessary) (Tenet 1). The phenomenon of people receiving a message and immediately assuming that they share the sender's worldview, values and experiences is known as "homophily" (Lazarsfeld and Merton, 1954). Since homophily does not automatically result in trust in an AI influencer but can be a significant factor leading to a high level of AI influence, antecedents such as physical attractiveness are unlikely to be enough to produce a high score for trust in an AI influencer (Selim et al., 2022; Kim and Kim, 2021; Youssef et al., 2022). The second tenet states that a high-quality result can be achieved through a combination of several different elements. The results show that it is possible to achieve high scores of trust in an AI influencer with only a

complicated configuration that derives from a combination of antecedents (physical attractiveness, homophily, congruencies, expertise and authenticity). Tenet 3 is the equifinality principle, which states that an outcome can occur without a proper model. The equifinality principle states that many solutions or paths to a conclusion may exist, depending on the antecedents (Shaalan et al., 2022; Woodside et al., 2015). The current research suggests that a high level of trust in an AI influencer can be earned through the use of various patterns forming a variety of factor combinations. The fourth tenet emphasises the concept of causal asymmetry, which holds that the causal recipes used to forecast trust in AI influencers are distinct and do not correspond to the inverse of any other consequence.

High scores on each recipe, which has its own special combination of components, correlate to similar results (Tenet 5). Trust in AI influencers, for instance, may be achieved by a combination of factors, including high physical attractiveness, homophily and high expertise. Furthermore, a high level of physical attractiveness, homophily, absence of expertise and strong congruencies all lead to the same high score for trust in AI influencers. The sixth tenet questions the conventional wisdom that X has a positive or negative correlation with Y. This tenet shifts the question from whether to when: in what circumstances does X have a positive or negative effect on Y? Trust in AI influencers' scores are more likely to be high if a specific causal configuration holds true. Asymmetric approaches take account of specific instances, while symmetric approaches focus on the big picture. In addition, not all customers start off with the same level of pre-existing trust in AI influencers, which can have a significant impact on how this trust is shaped.

2.2. Artificial intelligence influencer

Brands are increasingly using artificial intelligence (AI) to automate service delivery (Chi et al., 2020; Kim and Kim, 2021) and there is a general trend towards synthetic advertising (Jin et al., 2019). As a result, we may soon see digital persons in stores, or at least conversational interfaces powered by artificial intelligence (Reis et al., 2020). To help brands efficiently manage their social media accounts, AI is already being employed to automatically create and respond to posts (Schomer, 2019). With the advent of AI in the realm of marketing, a new type of influencer has emerged: the AI influencer (Huang, 2022; Schouten et al., 2020). Table 1 summarises previous research on AI influencers and provides a context for our own investigation. An artificial intelligence influencer can amass a sizable fan base and be seen as "a trusted taste-maker in one or more domains" (de Veirman et al., 2017, p. 798). Studies of AI and ML have also revealed promising outcomes that could result from the use of AI as a cultural force. The line between human and robot behaviour is blurring, making it easier for a robot to exert power (Chance, 2020).

AI influencers may mimic human appearance and behaviour, which suggests that they may have an impact on customers analogous to that of A-list celebrities. Although studies of AI influencers are beginning to emerge (Casalo' et al., 2020), little is known about customers' responses to them. It is unknown whether consumers will have the same negative reaction to robots as they have shown in other scenarios (Breves et al., 2019), or if they will treat human-looking AI influencers as if these were human. To build customers' trust in AI influencers in the retail sector, we look into three configurations: source attractiveness, congruencies and source credibility. Drawing on relevant previous studies and complexity theory, this paper aims to explain how the combination of source attractiveness (e.g., physical attractiveness and homophily), congruencies (influencer, product, consumer) and source credibility (e.g., expertise and authenticity) explains the causal recipes for raising trust in an AI influencer on social media.

2.3. Source credibility

One definition of source credibility is the impression that an audience has of a speaker at any given moment (Andersen and Clevenger, 1963; Ki et al., 2020). According to Aristotle, a listener's propitious attitude to a speaker (the communicator's credibility) may affect the acceptability of the message being received (Ohanian, 1990). Generally speaking, there are two main types of source credibility: "speaker expertise and speaker intent" (Hovland et al., 1953). According to research by Hovland et al. (1953, p. 21), the "extent to which a communicator was seen to be a source of valid statements" is an indicator of their level of expertise. One can tell how knowledgeable a speaker is by how much they know, how much schooling and experience they have had and how trustworthy their opinions are, among other things (Li et al., 2021; Umeogu, 2012). The perceived quality and authenticity of information are further enhanced by the speaker's credibility, which strengthens the speaker's authority (Umeogu, 2012). Expertise as a credible source led to consensus across target audiences and a shift in attitudes as a result of marketing messages.

Consumers' trust in an online information source depends on their belief that the content presented there is objective, credible and grounded in verifiable facts (Pick, 2020). One's own subjective evaluation of the reliability of information is what constitutes its credibility (Martínez-López et al., 2020a,b; Vrontis et al., 2021a,b). This impression is influenced by the information's origin, its content and the individuals receiving it (Kim and Kim, 2021). Hence, customers become resistant to persuasion attempts if they do not view the message as credible (Koay et al., 2021; Leite and Baptista, 2022).

The present study demonstrates that authenticity is a useful operationalization of the goals of influencers. According to sociologists (Lehman et al., 2019), authenticity is "being honest to oneself or to

Table 1
Results from the literature review on AI influencer.

Authors	Research focus	Method	Key findings
Arsenyan and Mirowska (2021)	"Investigates virtual agents' similarity to humans in terms of behavior and how reactions differ among types of influencers".	"Data scraping of influencer posts and public comments".	"Human-like virtual influencer receive significantly lower positive reactions from consumers".
lock and Lovegrove (2021)	"Analysis of an AI influencer's communication to identify what makes her so appealing to her audience".	"Textual and sentiment analysis of public posting".	"The influencer's identity and messaging effectively integrate into a single strategic communication tool, whereby the influencer is the strategy and the message".
Thomas and Fowler (2021)	"Investigates the implications of using AI influencers and sets out to provide guidance for when controversies arise from AI influencers".	"Two experimental studies".	"AI influencers can produce positive brand benefits like those produced by human celebrity endorsers".
Sands et al. (2022)	"Investigates how consumers respond to social media influencers that are created through artificial intelligence (AI) and compares effects to traditional (human) influencers".	"Two experimental studies".	"Compares AI and Human influencers to explore consumer perceptions, and finds similarities (following intentions, perceived personalization) and differences (AI lowers source trust and increases WOM). These effects are mediated by social distance and moderated by need for uniqueness and influencer agency".

others," which connotes "sincerity, genuineness, truthfulness and originality" (Van Leeuwen, 2001). In the view of Baker and Martinson (2002, P.17), "whether the practitioner is willing to openly, publicly and personally be identified as the persuader is a key indicator of authenticity in marketing communication". Since social media influencers are allowed to promote any product or service for financial gain, authenticity in this context refers to the influencer's true motivations in making a recommendation (Södergren, 2021). Users of social media platforms are wise to the practice of sponsorship and may view an influencer's justification for endorsing a product or service with scepticism. Hence, sincerity is required, to ensure that the AI influencer's audience will believe that the content that the influencer provides comes from the influencer's honest perspective. The present study suggests that expertise and authenticity improve customers' trust. Specifically, an influencer's competencies, expertise and competency all play a role in persuading their followers to trust the AI influencer (Lou and Yuan, 2019; Wu et al., 2022).

2.4. Source attractiveness

A source boosts the efficacy of its communication if it can attract much interest and participation (Park and Lin, 2020; Schimmelpfennig and Hunt, 2020). People are more likely to warm to and embrace the message of an attractive speaker. In McGuire's (1968) source attractiveness model, these qualities are taken into account, along with others like familiarity, resemblance, likeability and attractiveness. This model in the field of influencer marketing takes physical attractiveness and homophily to their source as criteria for source attractiveness.

Prior research revealed that homophily has a significant impact on the propensity to solicit feedback from and connect with others via social media (Kim and Kim, 2021). Homophily has been linked to increased curiosity, networking with key opinion leaders and knowledge sharing. Perceived homogeneity also boosts trust in the community and the reliability of news sources (Leite and Baptista, 2022). As a result, academics employ homophily to clarify how fans respond to material produced by celebrities, public figures and other types of "influencers" (Martínez-López et al., 2020a,b). The likelihood of interaction between followers and influencers can improve when they share similar traits (Koay et al., 2021). Surprisingly, on social media, it has been found that the similarity between the source of information and the receiver promotes interpersonal relationships and achieves connections and interactions (Pick, 2020), which in turn facilitates followers' engagement (Moriuchi, 2019) in the form of participatory behaviour (Gupta et al., 2022). To further boost the likelihood of influencing consumer behaviour, likeability owing to similarity also increases the notion that comparable followers are, by default, pleasant, kind and courteous (Ladhari et al., 2020).

Initial judgements and subsequent agreements are influenced by the communicator's physical beauty as evaluated by the receiver (Masuda et al., 2022). Positive traits, like kindness, interest, sociability, strength, modesty and receptivity, are attributed to those who are physically attractive (Dion et al., 1972). These people are believed to be enthusiastic, confident, knowledgeable and rational (Dipboye et al., 1977). When people form a favourable impression of the person communicating, they are more likely to believe what this person is saying and to share it with others. This is known as "homophily" (Lazarsfeld and Merton, 1954). Individuals who are able to connect with their audience are powerful influencers and can have a significant impact on the recommendations received by the audience (Masuda et al., 2022). Feelings of camaraderie and safety within a group are boosted and anxiety is lowered by a shared sense of identity (Kim and Kim, 2021). Listeners will probably agree with a speaker's statements if they infer that the speaker shares their outlook, interests, beliefs and emotions.

2.5. Congruencies

Buying habits can reveal a good deal about people, their values and their social circles (Casalo et al., 2020; Lee et al., 2012). In order to feel more at ease with their purchases, consumers often choose brands or items shown in ways that are consistent with their own sense of identity, a phenomenon known as "consumer-product congruence" (Fennis and Pruyn, 2007). With the help of theories such as balance (Belanche et al., 2021), cognitive dissonance (Jeong and Shin, 2020) and congruity (Park and Yoo, 2020), we further define the connections between influencers, sponsored products and fans (possible buyers). The nature of the connection between the customer and the influencer necessitates a high degree of fit between the two. Since the goal of native advertising is to be mistaken for journalistic material rather than overt promotion, consistency is crucial to eliciting positive responses from consumers. Internet visitors may be more receptive to native advertising if they believe it is organic and fits in with the surrounding material (Kim and Kim, 2021). For example, previous studies have shown that the advertising and product evaluation outcomes are more favourable when native advertisements are contextually matched (as opposed to less well matched) with a social media platform (Vrontis et al., 2021a,b). That is, customers have complete control over what they follow and unfollow at any given moment, giving them access to thousands of influencers over a wide range of topics (Swant, 2016). When clients identify with the idealised version of a public figure or find that their interests align with those of the figure's followers, a good connection is formed (Kim and Kim, 2021). Then, in order to avoid dissonance or imbalance, it is important for an influencer to recommend a product that is consistent with the image that the client wants to convey (Casalo' et al., 2020) and encounters on Instagram. In other words, if the influencer and the product are a good fit, the consumer, too, should have a favourable opinion of the product (Xu and Pratt, 2018).

According to a recant examination by Jia et al. (2021) that evaluated hotel robots with varying degrees of human similitude, consumers are less receptive to highly anthropomorphic robots, due to safety concerns and the simulation of human behaviour. Moreover, research by Sands et al. (2022) shows that people tend to have less trust in AI influencers on social media. Yu (2020) added that the use of humanoid robots in hotels could prompt negative feelings such as unease and insecurity. These disparate results point to the complexities of consumer behaviour when dealing with a novel AI influencer on social media, highlighting the need to take into account both the design/characteristics of the AI influencer and the features of the target customers (Zhu and Chang, 2020). Since customers tend to weigh the relative importance of numerous causal factors rather than a single one (Chuah and Yu, 2021), a more nuanced method of investigation is required. A configural examination of components, then, is more appropriate than an evaluation of individual causal elements for explaining and better understanding consumer trust in an AI influencer on social media. More nuanced causal patterns and more sophisticated interplay between the variables are revealed by adopting this viewpoint, as seen in Fig. 1.

The principle of equifinality is central to complexity theory. It states that any given event can be equally described by other sets of causal conditions that combine in sufficient configurations to produce the outcome in question (Fiss, 2007; Woodside et al., 2015). Customers' susceptibility to AI influencers on social media can be understood by considering configurations of source attractiveness, congruencies and source credibility. Customers are more likely to trust AI influencers on social media if they perceive the influencer to be physically appealing, homophilic, knowledgeable and sincere (Masuda et al., 2022). However, Kim and Kim (2021) find that, while customers' trust in social media influencers is affected by expertise and homophily and while authenticity affects customers' trust in influencers on social media, physical attractiveness has no significant effect on it.

Complexity theory also takes account of causal asymmetry. A condition's existence or absence in contributing to an outcome depends on

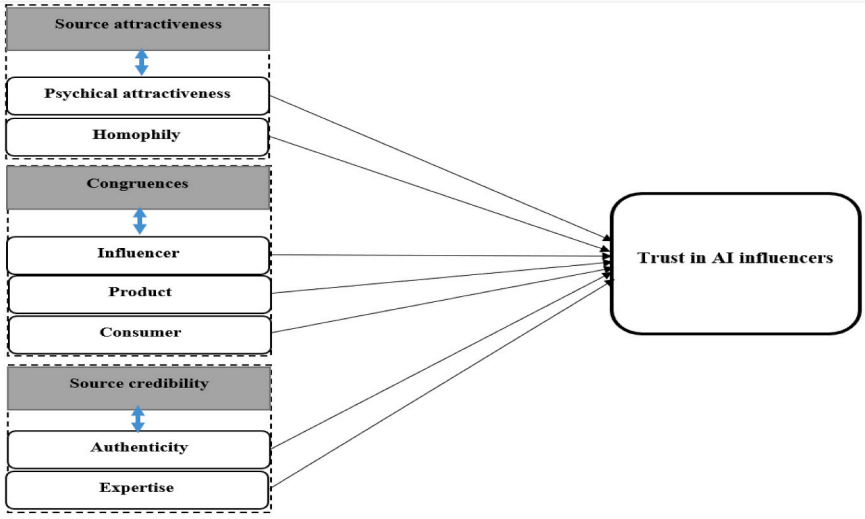


Fig. 1. Conceptual model.

how it combines with other circumstances, a phenomenon known as "causal asymmetry" (Woodside et al., 2015). High levels of trust may be attained by both customers with high levels of physical attractiveness and customers with low levels of physical attractiveness, depending on how they combine with the rest of the source attractiveness and source credibility (e.g., Kim and Kim, 2021; Masuda et al., 2022; Sands et al., 2022). Furthermore, source attractiveness or source credibility may represent alternate configurations of trust in AI influencers. For this reason, the following propositions are advanced in this research.

Proposition 1. *"No single best configuration of source attractiveness, congruencies and source credibility leads to high trust in AI influencers, but multiple, equally effective configurations of causal factors do exist"*.

Proposition 2. *"Single causal conditions may be present or absent within configurations for customers' high trust in an AI influencer, depending on how they combine with other causal conditions"*.

Proposition 3. *"Configurations that lead to high trust in an AI influencer will require the presence of at least one causal condition: source attractiveness, congruency or source credibility"*.

3. Methodology

3.1. Sampling and data collection

Our model was assessed through a quantitative approach taken to data gathered from a survey questionnaire based on a positivist research philosophy. The required data were gathered in August 2022. Some questions were included (i.e., "asking participants questions about their social media use and AI influencer following"). A brief definition of social media AI influencers was included to help respondents to understand the task. Respondents were asked to choose the AI which had influenced them most in the past six months. An established Saudi Arabian online survey company was used to collect data from consumers in Saudi Arabia (N = 683) who were already following an AI influencer on social media. Each consumer who took part was paid $1.50. Of these 683 participants, 361 were males (52.8%) and 322 females (47.2%). Participants reported using social media on a daily basis at a rate of 81.7% (n = 558), with 18.3% checking the influencer's posts daily. AI influencer content was found to be primarily focused on the following areas: fashion (31.6%), beauty (24.2%), travel (21.5%), technology and games (17.2%) (n = 116) and food (5.5%) (n = 39).

3.2. Measures

The measuring items regarding the proposed model for each construct were based on and refined from previous research-validated measures (Appendix 1). Expertise (e.g., "Experienced/Not experienced"), Authenticity (e.g., "Dependable/Not dependable") and Physical Attractiveness (e.g., "Attractive/Unattractive") were all measured using Kim and Kim's (2021) 5-point semantic differential scales. The following concepts were also measured using a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) based on the existing literature: homophily (e.g., "The source is similar to me in preferences and values") (Kim and Kim, 2021); trust (e.g., "The influencer can be relied upon in his (her) content") (Kennedy et al., 2001; Kim and Kim, 2021); and congruence among customer and influencer, adapted from Casalo' et al. (2020) (e.g., "this AI influencer is congruent with my values"). The concept of congruence between influencer and product was adapted from Xu and Pratt (2018) (e.g., "this AI influencer is a good match for the product"). Finally, the concept of congruence between customer and product was adapted from Xu and Pratt (2018) (e.g., "The product matches my style"). We used a back-translation method to translate the questionnaire into Arabic. We conducted a pilot assessment to assess the reliability and validity of the study instrument, testing it on 70 consumers who stated that they were familiar with the AI influencer. We used their feedback to refine the main instrument in terms of its readability, length, clarity and format. We also changed some of the wording in line with their comments.

3.3. Common method bias

Using principal component factor analysis, we were able to rule out the possibility of common method bias (Podsakoff et al., 2003). No single general construct explained more than 50% of the variance and the first (biggest) construct accounted for 34.07% (the variances explained a range from 16.39% to 34.07%), suggesting that common method bias may not be a severe issue in the data set. When applying the methodology proposed by Liang et al. (2021), we found that the substantive variance of indicators was 0.7, the average method-based variance was 0.007 and none of the method variable loadings was statistically significant. This led us to believe that the data set may have had a less than serious problem of common method bias. As a result, we could argue that common method bias may not be as significant an issue in the data set as we had at first thought.

4. Data analysis

4.1. Reliability and validity assessment

Before moving on to further analyses, we conducted a confirmatory factor analysis (CFA) to evaluate the construct's psychometric qualities. The CR values for all factors were greater than 0.70, indicating sufficient internal consistency, as determined by applying the composite reliability (CR) criterion to the factors (Hair et al., 2019) (see Table 2). In addition, since all of the outer loadings were greater than 0.794, we knew that the items could be trusted. Next, we validated the data by comparing the extracted variance to the sample mean (AVE). All of the factors' AVEs were greater than the minimum requirement of 0.5. Thus, we could infer that the factors' convergent validity was adequate. Using the HTMT method, we also tested the ability to discriminate (Henseler et al., 2016). Table 3 shows that all the HTMT ratios were significantly lower than 0.85, showing that the test revealed discriminant validity. The correlations between the study factors are indicated in Table 4.

The effect size *f2*, defined by Cohen (1988) as "the degree to which the phenomenon is present in the population," was also employed to investigate the substantive impact of the research design. Our model, following the advice of Cohen (1988), predicted a sizable effect from the trust-in-AI influencer ($f2 = 0.63$).

Armstrong and Overton (1977) established a strategy for evaluating the possibility of nonresponse bias, and we followed their lead by seeing if there were statistically significant differences between the early and late responders. In a chi-square analysis, there were no differences between early and late respondents at the 5% level of significance. Thus, nonresponse bias was ruled out.

All of the constructs were subjected to multivariate normality tests to ensure they met the criterion of skewness, kurtosis, and Mahalanobis distance statistics (Bagozzi and Yi, 1988). (Table 2). These showed no abnormalities at all.

4.2. Fuzzy-set qualitative comparative analysis (fsQCA)

In recent years, fuzzy set qualitative comparative analysis (FsQCA) has been increasingly popular in marketing research due to its versatility as a tool for combining the best features of qualitative and quantitative approaches (e.g., Beynon et al., 2020; Douglas et al., 2020). In contrast to the standard statistical method, which seeks to determine the net effects of a single variable, this method seeks to generate hypotheses that imply a relationship between several independent variables (Ragin and Fiss, 2008). This means that the result could be contingent on a number of factors working together. Set theorists can benefit from the configurational thinking approach since it is all-encompassing (Miller and Friesen, 1984), assuming equifinality and complicated causation (e.g., Fiss, 2007; Woodside, 2015), and enabling for the development of theoretical concept-based typologies (Yoruk and Jones, 2020).

Since this set-theoretic method makes it easier to establish causal connections, we used it to analyse the various possible configurations of the causal conditions (Ragin, 2008). In particular, QCA is used when trying to clarify complex causation and interactions between causal conditions (Fiss, 2007), under the assumption that one way to deal with complexity is to look for such permutations of elements as all add up to the same result. This method is preferred to the standard regression method, which looks at the "net effect" of factors (Pappas and Woodside, 2021). It is clear that the fsQCA is gaining popularity in a variety of business domains, particularly those dealing with Likert-scale data, because it is a QCA variant that uses membership degree assignment for

Table 2
Measurement statistics of construct scales.

Construct/Indicators	SFL	Mean	Standard deviation	Cronbach's α	CR	AVE	t-values	Skewness	Kurtosis
Trust in AI influencer (TRU)	0.939	2.834	1.127	0.931	0.950	0.592	23.209	−1.47	1.08
TRU1	0.908	3.038	1.038				15.234	−1.31	1.12
TRU2	0.941	2.991	1.320				12.093	−1.40	1.54
TRU3	0.973	3.007	1.175				27.120	−1.62	1.28
TRU4									
Psychical attractiveness (PSY)	0.920	2.631	1.478	0.937	0.962	0.618	21.094	−1.02	1.54
PSY1	0.945	3.029	1.239				19.783	−1.78	1.38
PSY2	0.919	2.672	1.301				20.472	−1.34	1.29
PSY3									
Homophily (HOM)	0.967	2.793	1.673	0.910	0.936	0.596	19.209	−1.05	1.21
HOM1	0.902	2.930	1.098				21.098	−1.20	1.20
HOM2	0.961	3.127	1.320				16.120	−1.45	1.56
HOM3									
Congruence between consumer and AI influencer (CCI)	0.940	2.304	1.430	0.947	0.962	0.593	18.037	−1.34	1.12
CCI1	0.939	3.478	1.209				21.209	−1.48	1.30
CCI2	0.908	2.992	1.438				13.028	−1.29	1.46
CCI3									
Congruence between AI influencer and product (CIP)	0.940	2.893	1.178	0.919	0.930	0.672	11.209	−1.43	1.28
CIP1	0.925	3.402	1.320				19.018	−1.12	1.09
CIP2	0.973	2.943	1.367				14.120	−1.20	1.27
CIP3	0.926	3.104	1.120				7.0392	−1.37	1.42
CIP4									
Congruence between consumer and product (CCP)	0.938	2.930	1.410	0.940	0.967	0.510	23.209	−1.45	1.29
CCP1	0.923	3.207	1.329				21.473	−1.04	1.10
CCP2	0.970	2.835	1.487				11.285	−1.38	1.43
CCP3	0.912	2.703	1.024				26.120	−1.20	1.25
CCP4									
Expertise (EXP)	0.946	3.803	1.217	0.938	0.970	0.562	12.120	−1.30	1.30
EXP1	0.907	2.620	1.045				19.403	−1.23	1.74
EXP2	0.910	3.102	1.178				9.028	−1.26	1.12
EXP3									
Authenticity (AUT)	0.907	2.893	1.120	0.905	0.928	0.690	14.203	−1.10	1.30
AUT1	0.932	3.100	1.328				11.209	−1.14	1.47
AUT2	0.912	2.327	1.127				21.438	−1.90	1.23
AUT3	0.946	2.885	1.214				17.034	−1.41	1.08
AUT4									

Notes: SFL: standardized factor loading; ****: SFL** is significant at the 0.001 level; **AVE** = Average variance extracted; **CR** = Composite reliability.

Table 3
Discriminant validity of the correlations between constructs.

Construct	Correlations and square roots of AVE							
	TRU	PSY	HOM	CCI	CIP	CCP	EXP	AUT
TRU	0.769							
PSY	0.493	**0.786**						
HOM	0.384	0.530	**0.772**					
CCI	0.321	0.418	0.561	**0.770**				
CIP	0.430	0.340	0.489	0.499	**0.819**			
CCP	0.458	0.425	0.336	0.502	0.510	**0.714**		
EXP	0.327	0.239	0.389	0.437	0.464	0.519	**0.749**	
AUT	0.454	0.402	0.451	0.482	0.488	0.326	0.416	**0.830**

a Composite reliability are along the diagonal, b Correlations.

Table 4
Correlations for model variables.

	1	2	3	4	5	6	7	8
(1) TRU	1.00							
(2) PSY	0.274**	1.00						
(3) HOM	0.308**	0.210**	1.00					
(4) CCI	0.195*	0.451**	0.219**	1.00				
(5) CIP	0.206**	0.165*	0.305**	0.430**	1.00			
(6) CCP	0.127*	0.108*	0.189*	0.316**	0.335**	1.00		
(7) EXP	0.419**	0.198**	0.104*	0.140*	0.297**	0.419**	1.00	
(8) AUT	0.227**	0.260**	0.143*	0.166*	0.157*	0.147*	0.229**	1.00

score scaling (Pappas and Woodside, 2021). Our research used the Quine-McCluskey method (included in the fsQCA 3.0 software package) to logically minimise the sufficiency claims, thereby reducing the complexity and yielding a more parsimonious solution.

4.3. Calibration

In the first step of fsQCA, scales are calibrated by converting the collected data into fuzzy sets, with membership ranging from "full non-membership" (i.e., 0) to "full membership" (i.e., 1), taking 0.5 as the "crossover point" (Ragin, 2008). The fsQCA software converts the original set of factors into a calibrated set by establishing three meaningful thresholds: "full membership, full nonmembership and the cross-over point", which characterises whether a given example is more in or more out of a given set (Ragin, 2008). Calibration was carried out as suggested by Pappas and Woodside (2021). The survey scale served as the basis for the three qualitative anchors used in this calibration method: "the full membership threshold, the full non-membership threshold and the crossover point threshold" (on 5-point Likert scales).

Table 5 indicates the filtered truth table (partial) with the outcome construct being trust in an AI influencer. The raw consistency explains the explanatory strength of the configuration with regard to the outcome construct. The analysis indicated that each predictor configuration could fully explain the generation of trust in the AI influencer. The results indicated no contradictory combination in the data (i.e., "the same antecedent condition combination produced different levels of results") and it was possible to perform subsequent comparative analysis.

4.4. fsQCA results

fsQCA3.0. was used for the purpose of data analysis. First, we built a truth table by ranking the frequency and consistency values to select an analytical configuration. We did this by ensuring that the remaining balance of configurations was valid more than 80% of the time and by adopting the minimal case frequency benchmark of 1 and the raw consistency benchmark of 0.8 (Pappas and Woodside, 2021; Ragin, 2008). Following these strict guidelines and filtering processes, we were able to derive the truth table rows necessary to move on to the next step and receive the findings for the configurational paths. The "Quine-McCluskey algorithm" was used to return the outcomes of three solutions:

Table 5
Truth table (partial).

PSY	HOM	CCI	CIP	CCP	EXP	AUT	Number	TRU	Raw consistency	PRI
0	1	1	0	1	0	1	81	1	0.948	0.637
1	1	1	0	1	0	1	23	1	0.935	0.729
1	1	0	1	1	0	1	11	1	0.956	0.734
1	1	0	1	1	1	0	11	1	0.937	0.719
0	1	1	1	1	1	1	4	0	0.966	0.735
0	1	1	1	1	1	1	5	1	0.941	0.761
1	1	1	1	1	1	1	7	1	0.936	0.720
0	1	1	1	1	1	1	21	1	0.955	0.693
0	0	1	1	1	1	1	8	1	0.959	0.679
1	1	1	1	1	0	1	34	1	0.938	0.682
1	1	1	1	0	0	1	6	1	0.933	0.714
1	0	1	0	1	1	1	6	1	0.970	0.662
1	0	1	0	1	1	0	7	0	0.951	0.729
1	1	1	1	1	1	1	9	0	0.943	0.694
1	1	1	1	1	1	1	5	1	0.952	0.619
0	1	1	1	1	1	1	12	1	0.940	0.644
0	1	1	1	1	1	1	18	1	0.936	0.683

namely the "parsimonious solution", "intermediate solution" and "complex solution". Due to its simpler assumptions and improved interpretability, we interpreted only the intermediate solution, following the advice of previous authors (Ragin, 2008). Pappas and Woodside (2021) proposed presenting both central and ancillary circumstances in each setup in order to gain a more nuanced understanding of the whole. In Table 5, the "black circles" (●) indicate the presence of a causal recipe, whereas the "crossed-out circles" (⊗) indicate the "absence or negation" of a causal recipe. The "blank squares" are understood to represent "do not care" situations, whereas the larger circles denote "core" requirements. Table 6 demonstrates that the fsQCA produced five solutions with a coverage of 0.893 and a consistency of 0.872. These five explanations thus account for a sizable portion of the outcome (i.e., trust in an AI influencer).

Table 6 shows the fsQCA results. The fsQCA function is grounded in the Quine-McCluskey theory of causation, which computes the causal recipes permitting the prediction of conditions that either increase or reduce trust in an AI influencer. Table 6 shows that a combination of source attractiveness, source credibility and congruencies result in five casual recipes leading to ahigh level of trust in an AI influencer (coverage = 0.795, consistency = 0.998). For instance, solution 1 indicates that high levels of physical attractiveness, homophily, congruencies between influencer and consumer and authenticity lead to high levels of trust in an AI influencer (coverage = 0.731, consistency = 0.963). Solution 2 revealed that a combination of physical attractiveness, homophily, expertise and authenticity can lead to high levels of trust in an AI influencer (coverage = 0.719, consistency = 0.970). Solution 3 indicated that the interaction between physical attractiveness, homophily, congruencies between influencer and product, expertise and authenticity lead to high levels of trust in an AI influencer (coverage = 0.680, consistency = 0.953).

Solution 4 shows that a combination of physical attractiveness, homophily and authenticity leads to high levels of trust in an AI influencer (coverage = 0.737, consistency = 0.982). Solution 5 demonstrates that a combination of physical attractiveness, homophily, congruencies between product and consumer and authenticity leads to high levels of trust in an AI influencer (coverage = 0.708, consistency = 0.981). In summary, the results of fsQCA indicated that no single factor was sufficient to drive trust in an AI influencer, but a combination of source attractiveness, source credibility and congruencies can lead to high levels of trust in an AI influencer. This result supports proposition 1. Moreover, the result indicated that alternative conditions might result in high levels of trust in an AI influencer. Thus, trust in an AI influencer can take different pathways. This result supports proposition 2, which suggested the occurrence of various predictors of trust in an AI influencer. Finally, when one driver condition interacts with another driving condition to produce a configurational cause, the results showed that asymmetrical effects arise. For example, configurations 1–5 include a high level of physical attractiveness and authenticity, while its negation is an ingredient in configurations 1 and 2. Proposition 3's prediction of the presence of asymmetrical effects is confirmed by these findings. Moreover, our study revealed two solutions that can lead to low level of trust in AI influencer. The first solution includes low levels of physical attractiveness, low homophily, low authenticity, and low expertise. The second configuration indicated that low homophily, low congruence among influencer and consumer, low authenticity, and low expertise can lead to low level of consumers trust toward AI influencer.

Table 6
Configurations for achieving high level of trust in AI influencer.

Configuration	Solution				
	1	2	3	4	5
Source attractiveness	●	●	●	●	●
Psychical attractiveness	●	●	●	●	●
Homophily	●	⊗	⊗	⊗	⊗
Congruencies	●	●	●	⊗	●
Consumer-AI influencer	⊗	●	●	●	•
AI influencer-product	0.963	0.970	•	0.982	⊗
Consumer-product	0.731	0.719	0.953	0.737	0.981
Source credibility	0.017	0.003	0.680	0.005	0.708
Authenticity	**0.998**		0.037		0.061
Expertise	**0.795**				
Consistency					
Raw coverage					
Unique coverage					
Overall solution consistency					
Overall solution coverage					

5. Discussion and implications

5.1. Key findings

Instagram has a growing number of individuals and economic volume (Statistica, 2022) and the diffusion of influencers on this visually appealing social networking site is on the rise (Kim and Kim, 2021; Sands et al., 2022), but little academic research has addressed trust in an AI influencer in the field of influencer marketing. Complexity theory and fsQCA were used to create and evaluate a model that details the main causal recipes that can produce a high level of trust in AI influencers. The major causal components that can lead to high levels of trust in AI influencers were discovered by combining source attractiveness, congruencies and source credibility.

Our study indicated five solutions that can lead to high levels of trust in AI influencers. The result confirms that physical attractiveness, homophily, congruencies between influencer and consumer and authenticity are key drivers of trust in an AI influencer. This finding is in line with source attractiveness and credibility frameworks (Hovland et al., 1953; McGuire, 1968). For instance, when an AI influencer is perceived as trustworthy and reliable, this may earn consumer trust (Giffin, 1967; Ohanian, 1990; Sternthal et al., 1978). McPherson et al. (2001) indicated that homophily is a key predictor of consumers' trust in an AI influencer. Our results contradict prior research findings that revealed that source attractiveness has no significant influence on consumer trust (Kim and Kim, 2021).

Previous studies have examined the independent influences of congruence between the influencers and the products (e.g., Kim and Kim, 2021; Sands et al., 2022; Xu and Pratt, 2018) or between the influencers and followers (e.g., Casalo' et al., 2020) and the findings emphasise that when influencer-customers' congruence is high, such influencer-product congruence encourages more consumer-product congruence. By examining the interconnected effects, we found that when the fans of an influencer view someone as congruent with their own values, personality, or image and this person promotes a product that seems to be congruent with the fans' usual style, the fans' perceptions of the product tend to align with the implied perceptions of the influencer. Specifically, they have a more favourable attitude to the product, which is consistent with the predictions of theories of cognitive dissonance (Greenwald and Ronis, 1978), balance (Heider, 1946) and congruity (Osgood and Tannenbaum, 1955). According to earlier studies (Belanche et al., 2021; Bu et al., 2022), including examinations involving influencers, this positive attitude then increases trust in the influencers (Kim and Kim, 2021).

5.2. Theoretical implications

Complexity theory is applied in this work to reveal the complexities of the relationships between AI influencers and followers, which makes a contribution to the literature on social media marketing. This research identifies five approaches that can be taken to increase consumers' faith in AI opinion leaders. This study creates a new perspective on influencer marketing by framing it as multi-layered relationship marketing on social media, drawing on insights from complexity theory that show how the successful exchange of resources stimulates actors to continue the

contact (Cartwright et al., 2022).

The results of the present study improve upon existing models of source credibility (Hovland et al., 1953) and appeal (Farivar et al., 2021; McGuire, 1968). This study indicates that the influencer's physical appearance significantly affected the trust in the AI influence that consumers felt. This discovery suggests that, over time, a social media influencer's physical appeal can help it gain and maintain its audience's trust. This result is best understood within the framework of interpersonal communication, and it lends credence to the claim that physical attractiveness can be conditionally effective (Maddux and Rogers, 1980). This study also indicated that the impact of expertise can be mitigated by strong personal connections. While the effect of expertise has been thought to be strong in the source credibility paradigm (Kim and Kim, 2021), this study indicated that expertise was important for increasing consumers' trust in AI influencers.

Our study explores an authentic AI influencer whose only claim to fame is her online activity, as opposed to a celebrity who is famous for reasons other than her social media presence (Hu et al., 2020). New types of opinion leader are emerging in the form of AI Influencers, who forge seemingly close bonds with their followers on the basis of common interests, values and tastes (Casalo' et al., 2020). Their proposals have weight because of the authority they have earned (Sands et al., 2022). Accordingly, the significance of congruence is expected to be amplified by the links between AI influencers and their followers, which offer greater insights into the lifestyle and interests of the AI influencer than traditional superstars make available. Understanding the nature of the content shared by AI influencers allows their followers to make more informed decisions about how to respond to promotional efforts. External validity is improved by analysing an AI influencer in practice.

This research is the first to examine the interplay between an influencer, a product and the end users in the context of influencer marketing. Influencer, product/brand and customers are the three primary players in every influencer marketing campaign (Stubb et al., 2019), so as much as possible should be learned about them. The impact of all the possible types of congruence between these three primary parts of an influencer campaign have not yet been determined, for previous research has analysed only some specific forms of congruence (e.g., "influencer-customer", Casalo' et al., 2020; "influencer-product", Kim and Kim, 2021). Previous studies (e.g., Casalo' et al., 2020; Kim and Kim, 2021) on how greater congruence between two parts of an influencer marketing campaign improves followers' responses are thus complemented by our findings. We demonstrate that increased success is possible when all three agents are congruent with one another. In this work, we contribute fresh insight by highlighting some interrelated effects: Together, congruences boost the effectiveness of AI-powered influencer marketing.

Our research makes important methodological progress by employing fsQCA as a theoretical framework for modelling trust in AI influencers. This method enables asymmetric modelling by delving into the factors that contribute to people's levels of trust in AI opinion leaders. We have shown, using this novel method, that a basic tactical plan is inappropriate because of the intricate interaction of numerous aspects, and that a really effective plan will need to take into account the level of sophistication with which people place their trust in AI influencers. To now, most research has only looked at how well indicators overall perform at predicting these sorts of behaviours. We have proven for the first time that a combination of source attractiveness, source trustworthiness, and congruences can lead to a high level of trust in AI influencers. Therefore, we have demonstrated the benefits of fsQCA over the regression-based technique, showing that survey data can be used in a large N QCA and providing evidence from more recent robustness tests to support this.

5.3. Managerial implications

Practical ramifications are also suggested by this study, especially for social media marketers who have lately been allocating more resources to AI influencer marketing and who are looking into methods for finding the most suitable influencer to work with (Ward and Gleditsch, 2018). Although influencers are appealing because of their many different qualities, marketers have found it hard to know how to choose the best influencer for their product (Kim and Kim, 2021). Instagram and Facebook, which provide user data and help those who are looking for influencers, can provide only demographic data and quantitative studies (Sands et al., 2022). This being the case, this research gives detailed criteria by which to evaluate an AI influencer's qualities, including knowledge, authenticity, physical attractiveness and homophily, in order to capitalise on relational trust.

Our study revealed that attractive characteristics and perceived similarity play important roles in establishing rapport and trust. We shed light on the contemporary phenomenon of "influencers," in which regular people have an immense effect on millions of their followers thanks to their approachable personas and contentment with life (Masuda et al., 2022). In order to capitalise on the trust established through interpersonal connections, marketers must decipher the similarities between influencers and their followers and apply that information to their campaigns. In addition, the findings suggest that AI influencer marketing is distinct from conventional celebrity endorsement, in which outward beauty plays a significant role. AI influencer marketing, in contrast to celebrity advertising, which seizes people's attention through the attractiveness of the people behind the brands, needs to figure out how to leverage physical attractiveness as a social resource for building relationships between influencers, consumers and companies.

Homophily, physical appeal and sincerity were all found to play a role in building trust, but the influencer's social attractiveness had the most impact. Researchers observed that homophily has a significant impact on trust creation in advertising across disciplines. An individual's admiration for influential people, such as celebrities, and/or the mental health of the followers of these people has been linked to the growth of trust in them. Thus, enhancing consumers' trust in video marketing using social media influencers can further optimise marketing performance by investigating how the key components affect the building of trust in the intended audience.

Influencer marketing strategies should always feel natural and in line with the influencer's personality and beliefs; the best results come when a product is seamlessly integrated into the influencer's life story (Casalo' et al., 2020). For this reason, businesses should seek to advertise their products in a way that is consistent with, or at least reminiscent of, the influencer's typical posting style. If the businesses succeed, consumers will have more good impressions and stronger intentions to purchase the sponsored product, due to the increased congruence between it and themselves.

6. Limitations and directions for future research

It should be noted that the current study has its caveats. In the first place, even though we think that the factors we found in this study helped us to understand trust in AI influencers, we acknowledge that certain other relevant factors which could affect the process were not included in our research. These factors include advertising literacy and followers' knowledge of the techniques of persuasion. Second, the study sought to ascertain the credibility of influencer-generated branded postings and the role that artificial intelligence influencers play in this context. But social media users' perceptions of influencers and the branded content generated by influencers may vary, depending on the platform they were using. This could be due to differences in message modality or influencers' content. AI-influenced marketing on particular SNSs may thus be an interesting subject for future research. Third, further examination may also wish to establish whether participants genuinely comprehend the notion of a social media AI influencer, even when they are given – as our participants were given – thorough information about the phenomenon of AI influencers on social media.

Additionally, a sample from Saudi Arabia was used to evaluate the models in this research. Future research into the impact of culture on social media engagement should be pursued.

While we think it's important to delve deeper into Instagram's analytics, other researchers may want to see if their findings hold up when applied to other social media platforms (like YouTube) or different industries (like the beauty or travel industries) or types of products and services (like cars, branded or unbranded products). In our empirical study, we use clothes, which can stand in for a variety of brands, as the "products." We did not include brand names to strengthen the experiment's internal validity and to prevent any familiarity or attachment effects. Any future research attempting to replicate our findings with brands should account for participants' pre-existing views and behavioural intentions toward the promoted companies. Finally, experimental designs and other methodologies can be used in future research to investigate the causal links between the study variables.

Declaration of competing interest

(None)

Data availability

Data will be made available on request.

Acknowledgements

(None)

Appendix 1. Variables' measurements

Trust in AI influencer (TRU) (Strongly Disagree (1)/Strongly Agree (5))		Source
TRU1	This AI influencer can be relied upon on his (her) content.	(Kennedy et al., 2001; Kim and Kim, 2021)
TRU2	I believe what this AI influencer says and that he/she would not try to take advantage of the followers.	
TRU3	This AI influencer is straightforward and honest even though his/her self-interests are involved.	
TRU4	This AI influencer would not tell a lie even if he/she could gain by it.	
Psychical attractiveness (PSY) (Strongly Disagree (1)/Strongly Agree (5))		**Source**
PSY1	This AI influencer is attractive.	Kim and Kim (2021)
PSY2	This AI influencer is beautiful.	
PSY3	This AI influencer is elegant.	
Homophily (HOM) (Strongly Disagree (1)/Strongly Agree (5))		**Source**
HOM1	This AI influencer has similar interests.	Kim and Kim (2021)
HOM2	This AI influencer has similar enthusiasm	
HOM3	This AI influencer have similar values.	
Congruence between consumer and AI influencer (CCI) (Strongly Disagree (1)/Strongly Agree (5))		**Source**
CC1	This AI influencer is congruent with my values.	(Casalo et al., 2020; Lee et al., 2012)
CC2	This AI influencer matches my personality.	
CC3	I feel identified with this AI influencer.	
Congruence between AI influencer and product (CIP) (Strongly Disagree (1)/Strongly Agree (5))		**Source**
CIP1	This AI influencer has a good match with the product.	(Belanche et al., 2021; Xu and Pratt, 2018)
CIP2	The compatibility between this AI influencer and the product is high.	
CIP3	The alignment between this AI influencer and the product is high.	
CIP4	This AI influencer and the product have a high fit.	
Congruence between consumer and product (CCP) (Strongly Disagree (1)/Strongly Agree (5))		
CCP1	The product matches my style.	(Belanche et al., 2021; Xu and Pratt, 2018)
CCP2	The compatibility between the product and me is high.	
CCP3	The alignment between the product bag and me is high.	
CCP4	The product fits my style.	
Expertise (EXP) (Strongly Disagree (1)/Strongly Agree (5))		**Source**
EXP1	I feel this AI influencer knows a lot.	(Lou and Yuan, 2019; Masuda et al., 2022)
EXP2	I feel this AI influencer is competent to make assertions about things that this AI influencer is good at.	
EXP3	I consider this AI influencer an expert on his/her area.	
EXP4	I consider this AI influencer sufficiently experienced to make assertions about his/her area.	
Authenticity (AUT) (Strongly Disagree (1)/Strongly Agree (5))		**Source**
AUT1	This AI influencer is dependable.	Kim and Kim (2021)
AUT2	This AI influencer is reliable.	
AUT3	This AI influencer is honest.	
AUT4	This AI influencer is trustworthy.	

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